Chapter 1

Introduction

1.1 Background

Border security and surveillance are important for protecting national boundaries and the safety of security personnel. Manual patrolling in dangerous areas can expose personnel to risks such as landmines and suspicious metallic objects. To reduce these risks, this project proposes a robot for border security and surveillance controlled remotely using an Android smartphone through Bluetooth communication.

The robot uses a microcontroller to control its movement, a wireless camera to monitor the surrounding area, and a metal sensor to detect metallic objects. GPS and GSM modules help determine the robot's location and send location information to the user. This system aims to support safer, more efficient surveillance and reduce the need for direct human entry into potentially dangerous areas.

The people who patrol the borders are also in danger because they can be ambushed at any time. They have to be careful about people shooting at them from the side of the border, and they have to watch out for bombs that are hidden. The weather is also a problem because sometimes there are big dust storms or it is very foggy, and it is hard to see anything. At night, it is also very dark and hard to see, which makes it easy for people to sneak across the border without being seen. Border security forces have to deal with these problems every day.

1.2 Problem Statement

Border security is an important task that requires continuous surveillance, monitoring, and detection of potentially dangerous objects. Manual patrolling in border areas can expose security personnel to risks such as landmines, suspicious metallic objects, and other hazardous situations. Traditional surveillance methods may also make it difficult to monitor certain areas safely and efficiently.

Therefore, this project aims to develop a robot for border security and surveillance that can be remotely controlled using an Android smartphone through Bluetooth communication. The robot is equipped with a wireless camera for real-time monitoring, a metal sensor for detecting metallic objects, and GPS/GSM modules for tracking and transmitting its location. The proposed system helps security personnel monitor potentially dangerous areas remotely, identify possible hazards, and reduce direct human exposure to risky environments.

The main challenge is to integrate mobility, sensing, communication, and monitoring capabilities into a reliable and cost-effective robotic platform. The system should be capable of operating in different environmental conditions while providing timely and useful information to the operator.

Therefore, this project aims to develop a robotic surveillance and monitoring system that improves situational awareness, reduces direct human exposure to hazardous environments, and supports safer and more efficient surveillance operations in modern warfare.

1.3 Objectives

1.3.1 Main Objective

The main objective of the project is to design and develop a robot for border security and surveillance that can be remotely controlled using an Android smartphone. The robot uses a wireless camera for monitoring, a metal sensor for detecting metallic objects, and GPS/GSM modules for location tracking and alert communication. It aims to improve security while reducing the risk to human personnel.

1.3.2. Specific Objectives

1. To develop a mobile robot capable of navigating different terrains and performing surveillance operations in potentially hazardous environments.
2. To monitor the surrounding area using a wireless camera and provide visual information to the operator.
3. **To detect metallic objects and potential landmine-related hazards using a metal sensor.**
4. **To enable real-time data transmission:** To establish wireless communication between the robot and the control station for transmitting video, sensor readings, and other monitoring data.
5. To reduce direct human exposure to potentially dangerous areas by enabling remote surveillance and detection operations.
6. To enable the robot to monitor locations that may be unsafe or difficult for human personnel to access directly.
7. To develop a robotic system that reduces the need for direct human exposure to dangerous environments and supports informed decision-making through timely surveillance data.

Chapter 2

Literature Review

Sr. No.	Author	Year	Key Findings
1	Ghareeb et al.	2017	The proposed system supports remote monitoring and transmits detected object locations and images, potentially improving operator safety and scanning efficiency.
2	Craig	2005	Robotics can improve speed, accuracy, flexibility, and efficiency while performing repetitive or hazardous tasks.
3	Li et al.	2015	The simulation results indicate that the Amorphous algorithm achieves the lowest positioning error under the evaluated conditions.
4	Magrabi et al.	2013	The study identifies safety risks and gaps in regulatory oversight, emphasizing the importance of standardization and system reliability.
5	Pugh and Martinoli	2007	The approach provides a framework for studying cooperative search behavior and the influence of communication range on multi-robot performance.
6	Rjeib et al.	2019	The system automates identification and record management, reducing manual documentation and improving information accessibility.
7	Suresh et al.	2015	The proposed robot supports eight-direction movement and remote operation and is intended to work on uneven surfaces.
8	Ambruš et al	2016	The study demonstrates that target shape can be estimated from limited spatial data despite uncertainties in sensor positioning and coil geometry.

9	Albert et al.	2014	The prototype demonstrates the potential of remote metal detection and solar energy for reducing dependence on external electrical supplies.
10	Ali et al.	2018	The system enables remote access to sensor readings and demonstrates real-time data transmission through a networked monitoring platform.
11	Zhao and Hwang	2023	Presented a ROS-based autonomous navigation robot platform using a stepping motor. The work relates to navigation platform design and motor-control integration.
12	Phueakthong and Varagul	2021	Developed a mobile robot using ROS 2 for navigation. The study provides a reference for implementing ROS 2-based navigation functions.
13	Borse et al.	2024	Presented a ROS 2-based navigation and simulation stack for the Robotino platform. The work is relevant to simulation-supported development and testing.
14	González-Santamarta et al.	2025	Introduced VAULT, a mobile mapping system for ROS 2-based autonomous robots. The study relates to mapping and navigation-system integration.
15	Sobhan et al.	2024	Analyzed timing and priority-driven enhancements for ROS 2 multithreaded executors. The work highlights performance considerations for time-sensitive robotic software.
16	Patel et al.	2022	Examined security vulnerabilities and forensic investigation in a ROS 2 case study. The findings underline the need to consider cybersecurity in connected robotic systems.
17	Mayoral-Vilches et al.	2022	Presented SROS2 cybersecurity tools for ROS 2. The work is relevant to protecting robot communications and improving system security.

18	Soriano et al.	2024	Discussed implementation of a Robot Intrusion Prevention System for ROS 2. The study provides context for intrusion monitoring and defensive software practices.
19	Kaur et al.	2024	Presented RoboCop, a framework for detecting zero-day cyber-physical attacks on robots. The work highlights the importance of monitoring abnormal behavior in robotic systems.
20	Macenski et al.	2023	Surveyed modern robotics algorithms available in ROS 2. The study provides an overview of algorithms that may support navigation and other robot functions.
21	Chen et al.	2024	Reviewed sensor technology for mobile robot navigation. The work supports consideration of suitable sensors for localization, obstacle awareness, and movement.
22	Authors not specified in the supplied reference	2024	Discussed indoor-environment surveillance using an autonomous mobile robotic device. The topic is directly related to mobile surveillance and autonomous patrol.
23	Kostiv et al.	2025	Studied self-supervised real-time tracking of military vehicles in low-frame-rate UAV footage. The work provides context for visual tracking, although its aerial-video setting differs from a ground robot.
24	Xie et al.	2021	Investigated multi-object tracking using deep-learning ensembles for unmanned aerial system applications. The study is relevant to understanding object-tracking approaches in visual data.
25	Ahmad et al.	2023	Reviewed computer-vision techniques for military surveillance drones. The work provides background on vision-based surveillance methods and their application constraints.

26	Park et al.	2023	Presented multi-object tracking on SWIR images for city surveillance. The study illustrates tracking methods in specialized imaging conditions.
27	Karison et al.	2024	Analyzed errors in a controlled turret system using AI-based target-location input. This reference concerns targeting systems and is not directly required for a non-weaponized support robot.
28	Keswani	2025	Presented an AI sentry turret concept for area surveillance. The work concerns automated surveillance hardware; weapon-related functions should be excluded from a safe logistics/support robot.

Chapter 3

Components

- **RESISTORS :**



In many electronic circuit applications the resistance forms the basic part of the circuit. The reason for inserting the resistance is to reduce current or to produce the desired voltage drop . These components which offer value of resistance are known as resistors . Resistors may have fixed value i.e., whose value cannot be changed and are known as fixed resistors . Such of those resistors whose value can be changed or varied are known as variable resistors.

There are two types of resistors available. They are :

- Carbon resistors .
- Wire wound resistors .

Carbon resistors are used when the power dissipation is less than 2W because they are smaller and cost less. Wire wound resistors are used where the power dissipation is more than 5W . In electronic equipments carbon resistors are widely used because of their smaller size .

All resistors have three main characteristics:

- Its resistance R in ohms (from 1 ohm to many mega ohms).
- Power rating (from several 0.1W to 10 W) .
- Tolerance (in percentage) .

➢ RESISTOR COLOUR CODING

The carbon resistors are small in size and are color coded to indicate their resistance value in ohms. Different colors are used to indicate the numeric values. The dark colors represent lower values and the lighter colors represent the higher values. The color code has been standardized by the electronic industries association.

The color bands are printed at one end of the resistors and are read from the left to right. The first color band closed to the edge indicates the first digit in the value of resistance .The second band gives the second digit. The third band gives the number of zero's after two digits. The resulting number is the resistance in ohms. A fourth band indicates the tolerance i.e., to indicate how accurate the resistance value is , the bands are shown in the figure 1.

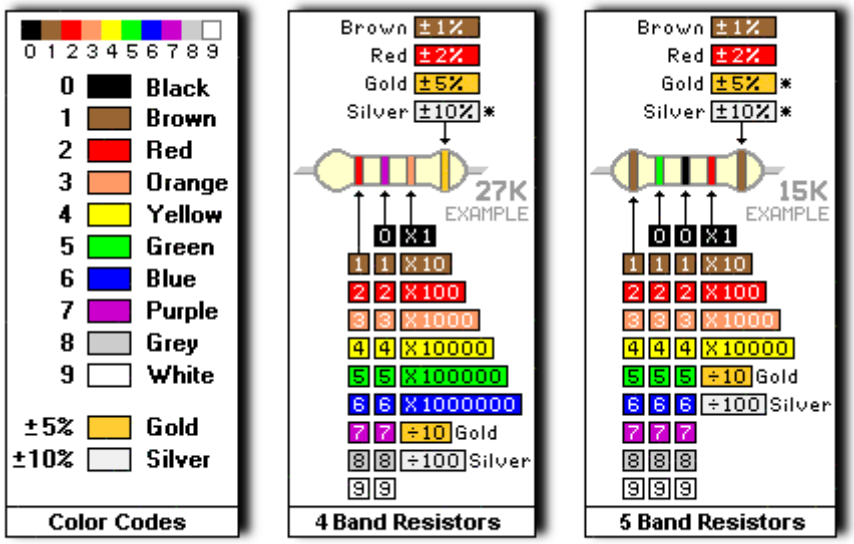


Fig. 1: Colour code for Resistor

- **PRESET**



There are two general categories of variable resistors:

- General purpose resistors.
- Precision resistors.

The general purpose type can again be wire wound type and carbon type .These follows either linear or logarithmic law. The precision type are always wire wound and follow a linear law .The variable resistors can be broadly classified as potentiometer , rheostats , presets and decade resistance boxes.

The general purpose wire wound potentiometers are available in 1, 2, 3 and 4 watts. The usual tolerances ratings 10 % and 20% are available. The widely used potentiometers are of the standard diameters 19mm, 31mm, and 44mm. The temperature coefficient depends on the wire used and on the resistors values. The resolution of these wire wound resistors is proper than carbon resistors because the wiper has to move from one winding to the other, where as in carbon potentiometers it is continuous. These resistors are highly linear, the linearity falling with 1%.

- **CAPACITORS:**

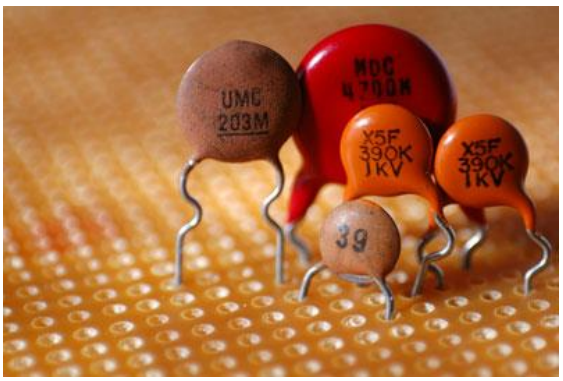
Devices which can store electronic charge are called capacitors. Capacitance can be understood as the ability of a dielectric to store electric charges. Its unit is Farad, named after the Michael Faraday. The capacitors are named according to the dielectric used.

Most common ones are air, paper, and mica, ceramic and electrolytic capacitors.

Physically a capacitor has conducting plates separated by an insulator or the dielectric. The plates of the capacitor have opposite charge, this gives rise to an electric field .In capacitor the electric field is concentrated in the dielectric between the plates.
Like resistors, capacitors are also crucial to the correct working of nearly every electronic circuit and provide us with a means of storing electrical energy in the form of an electric field. Capacitors have numerous applications including storage capacitors in power supplies, coupling of A.C. signals between the stages of an amplifier, and decoupling power supply rails so that, As far as A.C. signal components are concerned, the supply rails are indistinguishable from zero volts.

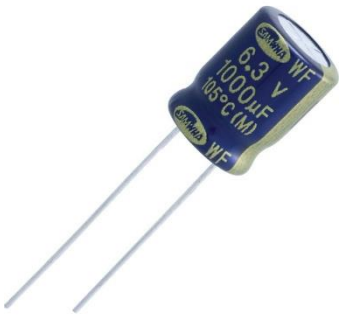
- **TYPES OF CAPACITORS**:

- **DISC CAPACITORS :**



In the disk form, silver is fired on to both sides of the ceramic to form the conductor plates. The sheets are then baked and cut to the appropriate shape and size & attached by pressure contact and soldering . These have high capacitance per unit volume and are very economical. The disks are lacquered or encapsulated in plastic or Phenolic molding. Round disk are used at high voltages the capacitance of values upto 0.01F can be obtained. They have tolerance of +20% or –20%. In general these capacitors have voltage ratings upto 750 V d.c.

- **ELECTROLYTIC CAPACITORS :**



These capacitors derive the name from electrolyte which is used as a medium to produce high dielectric constants. These capacitors have low value for large capacitances at low working voltages.

There are two types of Electrolytic capacitors:

- Aluminum Electrolytic capacitors .
- Tantalum electrolytic capacitors .

Electrolytic capacitors are used in circuits that have combination of D.C. voltage and A.C. The D.C. voltage maintains the polarity . They are used as 'ripple filter ' where large capacitance are required at low cost in small space . They are also used as 'biased capacitors ' and 'decoupling capacitors ' and even as 'coupling capacitors ' in R- C amplifier.

- **COLOUR CODING :**

Mica and tubular ceramic capacitors are color coded to indicate a capacitance value . As coding is necessary only for very small sizes, color coded capacitors value is also in the pF . The colors are the same as for the resistor coding from black for '0' upto white for '9'. Mica capacitors use ' six dot code system'.

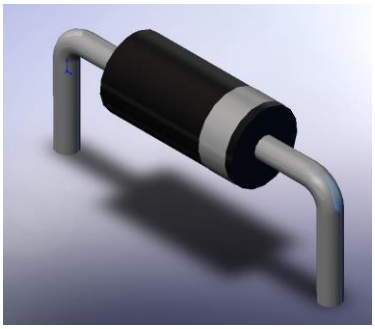
- **SIX DOT CODE :**

Here the top row is read from the left to right and the bottom from right to left .The dot indicates the following:

(1). White. (2). Digit. (3). Digit. (4). Multiplier. (5). Tolerance. (6). Class.

White for the first dot indicates the coding. The capacitance value is read from the next three dots. If the first dot is silver it indicates paper capacitor. The white colored band indicates the left and specifies the temperature coefficient . The next three colors indicate the value of capacitance . For example Brown, Black, Brown = 100 pF.

DIODES:



To ensure unidirectional flow of liquid we use mechanical valves in its path. By properly arranging these valves in a system we get useful devices such as pumps and locomotives. In the field of electronics too we have a valve called semiconductor diode (a counterpart of thermionic valve) for controlling the flow of electric current in one direction. But we use these diodes in circuits for limited purposes like converting AC to DC, by passing EMF etc. a diode allows current to pass through it provided it is forward biased and the biasing voltage is more than potential barrier (forward voltage drop) of the diode

- **AUTOMATIC SWITCHOVER TO BATTERY**

An uninterrupted power supply (UPS) is necessary for a main operated clock. This facility is very useful in transistors and two in ones for recording or listening to news programs. A relay can do this job with a battery backup. But the relay takes several milliseconds before it makes contact. Moreover, it is costly and occupies space.

The same task can be achieved with a single diode. Just connect a germanium diode DR50 (D1) as shown in fig 1.when the power is available form the eliminator or the external power source, the gadget will use the power from it. As points A and B are at same potential, the external power is remove, point B will be at higher potential that point A i.e.

D1 is forward biased and current flows from the battery. In no case the voltage of the eliminator or the external power source should be less than the voltage of the battery. Otherwise, the current will flow from the battery during mains operation also and the battery will be drained quickly.

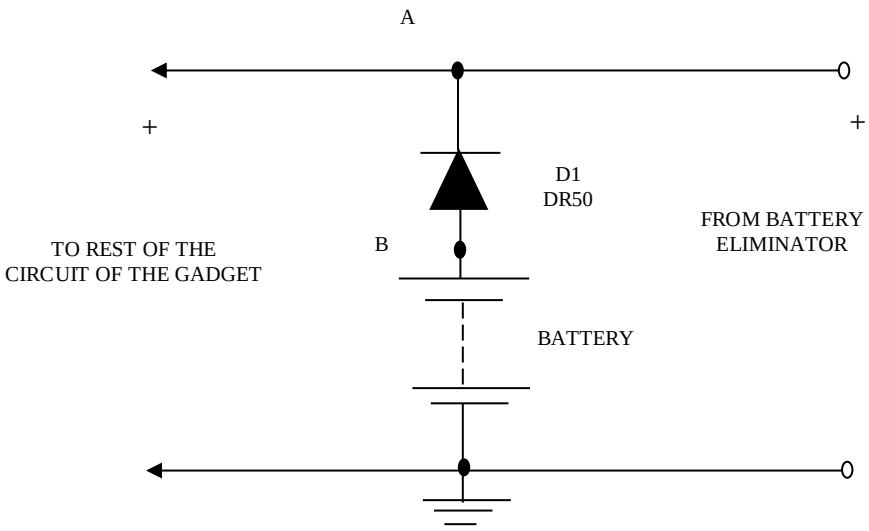
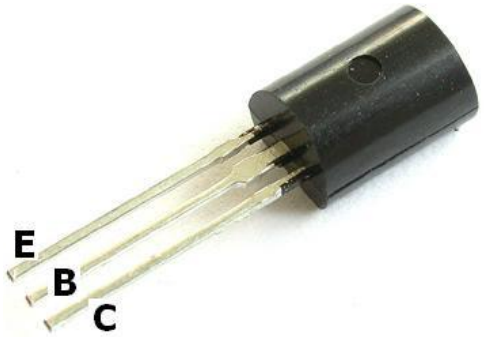


FIG 1: AUTOMATIC SWITCHOVER TO BATTERY

- **TRANSISTOR:**



The transistor an entirely new type of electronic device is capable of achieving amplification of weak signals in a fashion comparable and often superior to that realized by vacuum tubes. Transistors are far smaller than vacuum tube, have no filaments and hence need no heating power and may be operates in any position. They are mechanically strong, hence practically unlimited life and can do some jobs better than vacuum tubes.

Invented in 1948 by J. Bardeen and W.H.Brattain of Bell Telephone Laboratories, a

transistor has now become the heart of most electronic appliance. Though transistor is only slightly more the 45 years old, yet it is fast replacing vacuum tubes in almost all applications.

A transistor consists of two pn junction formed by sand witching either p-type or n-type semiconductor between a pair of opposite type. Accordingly, there are two types of transistors namely:

- n-p-n transistor
- p-n-p transistor

An n-p-n is composed of two n-type semiconductors separated a by thin section of p-type. However, a p-n-p is formed by two p-section separated by a thin section of n-type.

- These are two pn junctions. Therefore, a transistor may be regarded as a combination of two diodes connected back to back.
- There are 3 terminals, taken from each type of semiconductor.
- The middle section is very thin layer. This is the most important factor in the functioning of a transistor.

Origin of the name "transistor ": When new devices are invented, scientists often try to device a name that will appropriately describe the device. A transistor has two pn junctions. As the discussed later one junction is forward biased and the other is reversed biased. The forward biased junction has low resistance path whereas the reverse biased junction has low resistance path whereas the reverse biased junction has a high resistance path. The weak signal is introduced in the low resistance circuit and output is taken from the high resistance circuit. Therefore, a transistor transfers a signal from a low resistance to high resistance. The prefix 'tans' means the signal transfer property of the device while 'istor' classifies it as a solid element in the same general family with resistors.

➢ NAMING THE TRANSISTOR TERMINALS:

A transistor (PnP or NPN) has three sections of doped semiconductors. The section on one side is the emitter and the section on the opposite side is the collector. The middle section is called the base and forms two junctions between the emitter and collector.

- ❖ Emitter: - The section on one side that supplies charge carriers (electrons or holes) is called the emitter. The emitter is always forward biased w.r.t base so that it can supply a large number of majority carriers.
- ❖ Collector: - The section on the other side that collects the charge is called the collector. The collector is always reversing biased. Its function is to remove charges from its junction with the base.
- ❖ Base: - The middle section, which forms to pn junctions between the emitter and collector, is called base. The base emitter junction is forward biased, allowing low resistance for the emitter circuit. The base-collector junction is reversed biased and provides high resistance in the collector circuit.

➢ CHARCTERISTICS OF TRANSISTORS

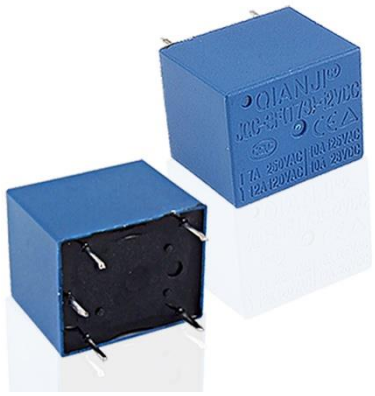
Whenever we have to decide about the applications of a transistor certain question arises. Some of these are – how much amplification gets from it? What is the highest frequency upto which it can be used? How much power output could we get from it? And what should be the values of different components used in the circuits? The answers to these entire questions lie in the electrical properties of the transistor. These properties depend on the size, manufacturing techniques and materials used in the

manufacturer of transistor and are know as characteristics. Transistor manufacturers give these characteristics in the data sheets published by them.

- ❖ Current gain factor 'alpha' (α)
- ❖ Current gain factor 'beta' (β)
- ❖ Input resistance (Rin)
- ❖ Output resistance (Rout)
- ❖ Cut-off frequency (F α and Fβ)
- ❖ Leakage current (I 'co)
- ❖ Maximum collector voltage (V_{ceo})

- ❖ Maximum emitter current (IC Max)
- ❖ Maximum Power dissipation (P max)

➢ RELAY:



A **relay** is an electrically operated switch. Many relays use an electromagnet to operate a switching mechanism mechanically, but other operating principles are also used. Relays are used where it is necessary to control a circuit by a low-power signal (with complete electrical isolation between control and controlled circuits), or where several circuits must be controlled by one signal. The first relays were used in long distance telegraph circuits, repeating the signal coming in from one circuit and re-transmitting it to another. Relays were used extensively in telephone exchanges and early computers to perform logical operations.A type of relay that can handle the high power required to directly drive an electric motor is called a contactor. Solid-state relays control power circuits with no moving parts, instead using a semiconductor device to perform switching. Relays with calibrated operating characteristics and sometimes multiple operating coils are used to protect electrical circuits from overload or faults; in modern electric power systems these functions are performed by digital instruments still called "protective relays".

➢ BASIC DESIGN AND OPERATION:

A simple electromagnetic relay consists of a coil of wire surrounding a soft iron core, an iron yoke which provides a low reluctance path for magnetic flux, a movable iron armature, and one or more sets of contacts (there are two in the relay pictured). The armature is hinged to the yoke and mechanically linked to one or more sets of moving contacts. It is held in

place by a spring so that when the relay is de-energized there is an air gap in the magnetic circuit. In this condition, one of the two sets of contacts in the relay pictured is closed, and the other set is open. Other relays may have more or fewer sets of contacts depending on their function. The relay in the picture also has a wire connecting the armature to the yoke. This ensures continuity of the circuit between the moving contacts on the armature, and the circuit track on the printed circuit board (PCB) via the yoke, which is soldered to the PCB.

When an electric current is passed through the coil it generates a magnetic field that attracts the armature and the consequent movement of the movable contact either makes or breaks (depending upon construction) a connection with a fixed contact. If the set of contacts was closed when the relay was de-energized, then the movement opens the contacts and breaks the connection, and vice versa if the contacts were open. When the current to the coil is switched off, the armature is returned by a force, approximately half as strong as the magnetic force, to its relaxed position. Usually this force is provided by a spring, but gravity is also used commonly in industrial motor starters. Most relays are manufactured to operate quickly. In a low-voltage application this reduces noise; in a high voltage or current application it reduces arcing.

When the coil is energized with direct current, a diode is often placed across the coil to dissipate the energy from the collapsing magnetic field at deactivation, which would otherwise generate a voltage spike dangerous to semiconductor circuit components. Some automotive relays include a diode inside the relay case. Alternatively, a contact protection network consisting of a capacitor and resistor in series (snubber circuit) may absorb the surge. If the coil is designed to be energized

with alternating current (AC), a small copper "shading ring" can be crimped to the end of the solenoid, creating a small out-of-phase current which increases the minimum pull on the armature during the AC cycle.[1]

A solid-state relay uses a thyristor or other solid-state switching device, activated by the control signal, to switch the controlled load, instead of a solenoid. An optocoupler (a light-emitting diode (LED) coupled with a photo transistor) can be used to isolate control and controlled circuits.

- **DC MOTOR:**



The direct current (DC) motor is one of the first machines devised to convert electrical power into mechanical power. Permanent magnet (PM) direct current convert electrical energy into mechanical energy through the interaction of two magnetic fields. One field is produced by a permanent magnet assembly, the other field is produced by an electrical current flowing in the motor windings. These two fields result in a torque which tends to rotate the rotor. As the rotor turns, the current in the windings is commutated to produce a continuous torque output Permanent magnet (PM) motors are propably the most commonly used DC motors, but there are also some other type of DC motors (types which use coils to make the permanent magentic field also).

DC motors operate from a direct current power source. Movement of the magnetic field is achieved by switching current between coils within the motor. This action is called "commutation". Very many DC motors (brush-type) have built-in commutation, meaning that as the motor rotates, mechanical brushes automatically commutate coils on the rotor. Motor speed control of DC motor is nothing new. A simplest method to control the rotation speed of a DC motor is to control it's driving voltage. The higher the voltage is, the higher speed the motor tries to reach. In many applications a simple votlage regulation would cause lots of power loss on control circuit, so a pulse width modulation method (PWM) is used in many DC motor controlling applications. In the basic Pulse Width Modulation (PWM) method, the operating power to the motors is

turned on and off to modulate the current to the motor. The ratio of "on" time to "off" time is what determines the speed of the motor.

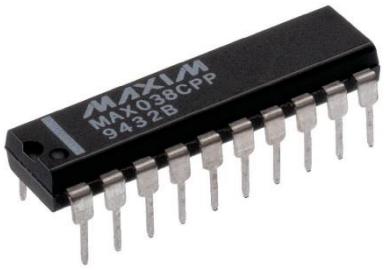
Sometimes the rotation direction needs to be changed. In normal prmanent magnet motors, this rotation is changed by changing the polarity of operating power (for example by

switching from negative power supply topositive or by interchanging the power terminals going to power supply). This directrion chanign is typicaly implemented using relay or a circuit called an H bridge.

Besides "brush-type" DC motors, there is another DC motor type: brushless DC motor. Brushless DC motors rely on the external power drive to perform the commutation of stationary copper winding on the stator. This changing stator field makes the permanent magnet rotor to rotate. A brushless permanent magnet motor is the highest performing motor in terms of torque / vs. weight or efficiency. Brushless motors are usually the most expensive type of motor. Electronically commutated, brush-less DC motor systems are widely used as drives for blowers and fans used in electronics, telecommunications and industrial equipment applications. There is wide variety of different brush-less motors for various applications. Some are designed to to rotate at constant speed (those used in disk drives) and the speed of some can be controlled by varying the voltage applid to them (usually the motors used in fans). Some brushless DC motors have a built-in tachometer which gives out pulses as the motor rotates (this applies to both disk drive motors and some computer fans).

In general, users select brush-type DC motors when low system cost is a priority, and brushless motors to fulfil other requirements (such as maintenance-free operation, high speeds, and explosive environments where sparking could be hazardous). Brush type DC motors are used in very many battery powered appliances. Brushless DC motors are commonly used in applications like DC powered fans and disk drive rotation motors.

- **INTRODUCTION TO INTEGRATED CIRCUITS**
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All modern digital systems rely on the use of integrated circuits in which hundreds of thousands of components are fabricated on a single chip of silicon. A relative measure of

the number of individual semiconductor devices within the chip is given by referring to its 'scale of integration'. The following terminology is commonly applied.

Scale of integration	Abbreviation	Number of logic gates
Small	SSI	1 to 10
Medium	MSI	10 to 100
Large	LSI	100 to 1000
Very large	VLSI	1000 to 10,000
Super large	SLSI	10,000to 100,000

- **ENCAPSULATION**

The most common package used to encapsulate an integrated circuit, and that with which most reader will be familiar, is the plastic dual-in-line (DIL) type. These are available with a differing number of pins depending upon the complexity of the integrated circuit in question and, in particular, the need to provide external connections to the device. Conventional logic gates, for example, are often supplied in 14-pin or 16-pin DIL packages, whilst microprocessors (and their more complex support devices) often require 40-pins or more.

- **IDENTIFICATION**

When delving into an unfamiliar piece of equipment, one of the most common problems is that of identifying the integrated circuit devices. To aid us in this task, manufacturers provide some coding on the upper surface of each chip. Such a coding generally includes the type number of the chip (including some of the generic coding), the manufacturer's name (usually in the form of prefix letters), and the classification of the device (in the form of a prefix, infix or suffix).

In many cases the coding is further extended to indicate such things as encapsulation, date of manufacture, and any special characteristics of the device. Unfortunately, all of this potentially useful information often leads to some considerable confusion due to inconsistencies in marking from one manufacturer to the next!

➢ LOGIC FAMILIES

The integrated circuit device on which modern digital circuitry depends belongs to one or other of several 'logic families'. The term simply describes the type of semiconductor technology employed in the fabrication of the integrated circuit. This technology is instrumental in determining the characteristics of a particular device. This, however, is quite different from its characteristics, and encompasses such important criteria as supply voltage, power dissipation, switching speed and immunity to noise.

The most popular logic families, at least as far as the more basic general purpose devices are concerned, are complementary metal oxide semiconductor (CMOS) and transistor transistor logic (TTL). TTL also has a number of sub-families including the popular low power Schottky (LS-TTL) variants.

The most common range of conventional TTL logic devices is known as the '74' series. These devices are, not surprisingly, distinguished by the prefix number 74 in their coding. Thus devices coded with the numbers 7400, 7408, 7432 and 74121 are all members of this family which is often referred to as 'Standard TTL'. Low power Schottky variants of these devices are distinguished by an LS infix. The coding would then be 74LS00, 74LS08, 74LS32 and 74LS121.

Popular CMOS devices from part of the '4000' series and are coded with an initial prefix of 4. Thus 4001, 4174, 4501 and 4574 are all CMOS devices. CMOS devices are sometimes also given a suffix letter; A to denote the 'original' (now obsolete) unbuffered series, and B to denote the improved (buffered) series. A UB suffix denotes an unbuffered B-series device.

Infix letters	Meaning
C	CMOS version of a corresponding TTL device
F	'Fast' – a high speed version of the device
H	High speed version
S	Schottky (a name resulting from the input circuit Configuration)
HC	High speed CMOS version (with CMOS compatible inputs)
HCT	High speed CMOS version (with TTL compatible inputs)

- Calibrate encoders and IMU; configure coordinate frames and sensor mounting carefully.
- Test sensor failure, communication loss and emergency-stop behaviour in a controlled test area.
- Use authenticated communication and do not rely only on wireless connectivity to stop the robot.
- Mapping and navigation are not automatic just because ROS 2 is installed; packages, configuration, calibration and testing are required.

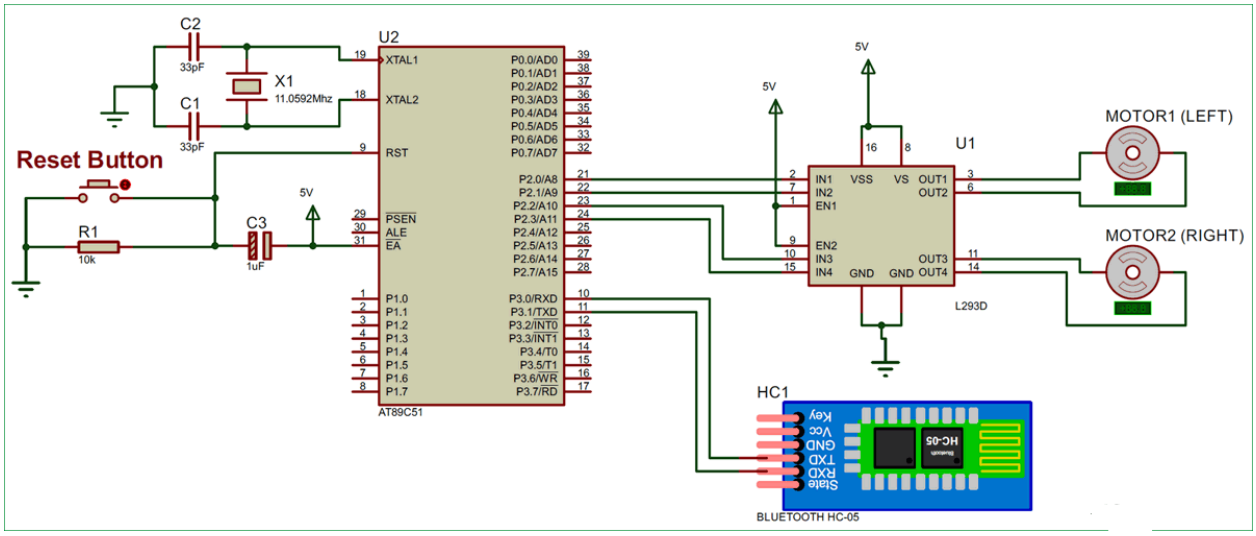
Chapter 4 :

Circuit Diagram & Explanation

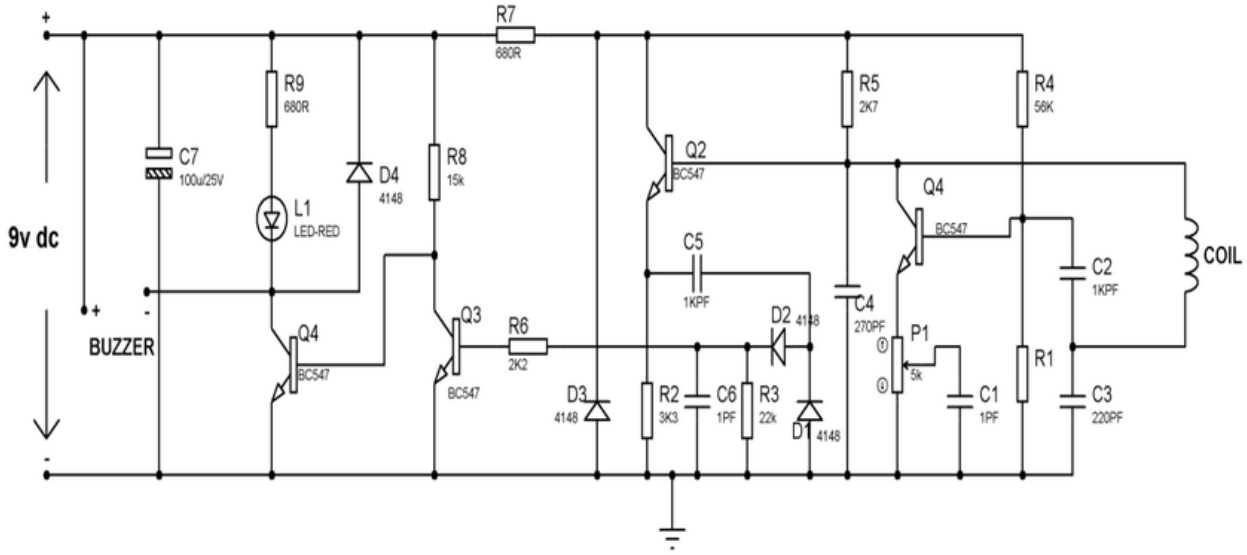
4.1Circuit Diagram & Explanation

A robot which moves freely in a building has to be adapted to an environment made for humans. This technology has applications as military robots or security robots in urban environments where security operation is an essential part of the mission. The robot is operates in terrain where wheeled robots would operate.

In this highly developing society; time and man power are critical constrains for completion of task in large scales. The automation is playing important role to save human efforts in most of the regular and frequently carried works e.g. most of the industrial/commercial/monitoring jobs like bomb detection, human detection, fire fighting etc. one of the major and most commonly performed surveillance. For this purpose, metal/bomb robot may be used which operator can control robot from safe distance for life saving purpose.



Circuit 1.



Circuit 2. Metal detector

This is a Beat Frequency based metal detector, which works by using the lower heterodyne of two mixed signals coming from two separate oscillators. One oscillator generates a stable reference frequency, while the other oscillator works as a VFO (Variable Frequency Oscillator). Typically, the two oscillators are in sync, and there is no heterodyne generation by their output mix. While the reference oscillator keeps the frequency stable, the VFO changes its frequency in the presence of a metal object.

- **POWER SUPPLY UNIT**

The circuit needs two different voltages, +5V & +12V, to work. These dual voltages are supplied by this specially designed power supply.

The power supply, unsung hero of every electronic circuit, plays very important role in smooth running of the connected circuit. The main object of this 'power supply' is, as the name itself implies, to deliver the required amount of stabilized and pure power to the circuit. Every typical power supply contains the following sections:

1. Step-down Transformer: The conventional supply, which is generally available to the user, is 230V AC. It is necessary to step down the mains supply to the desired level. This is achieved by using suitably rated step-down transformer. While designing the power supply, it is necessary to go for little higher rating transformer than the required one. The reason for this is, for proper working of the regulator IC (say KIA 7805) it needs at least 2.5V more than the expected output voltage

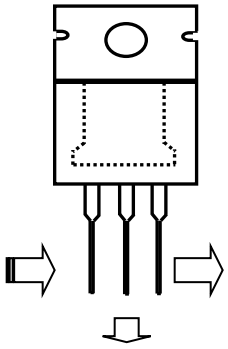
2. Rectifier stage: Then the step-downed Alternating Current is converted into Direct Current. This rectification is achieved by using passive components such as diodes. If the power supply is designed for low voltage/current drawing loads/circuits (say +5V), it is sufficient to employ full-wave rectifier with centre-tap transformer as a power source. While choosing the diodes the PIV rating is taken into consideration.

3. Filter stage: But this rectified output contains some percentage of superimposed a.c. ripples. So to filter these a.c. components filter stage is built around the rectifier stage. The cheap, reliable, simple and effective filtering for low current drawing loads (say upto 50 mA) is done by using shunt capacitors. This electrolytic capacitor has polarities, take care while connecting the circuit.

4. Voltage Regulation: The filtered d.c. output is not stable. It varies in accordance with the fluctuations in mains supply or varying load current. This variation of load current is observed due to voltage drop in transformer windings, rectifier and filter circuit. These variations in d.c. output voltage may cause inaccurate or erratic operation or even malfunctioning of many electronic circuits. For example, the circuit boards which are

implanted by CMOS or TTL ICs.

KIA 78xx Series

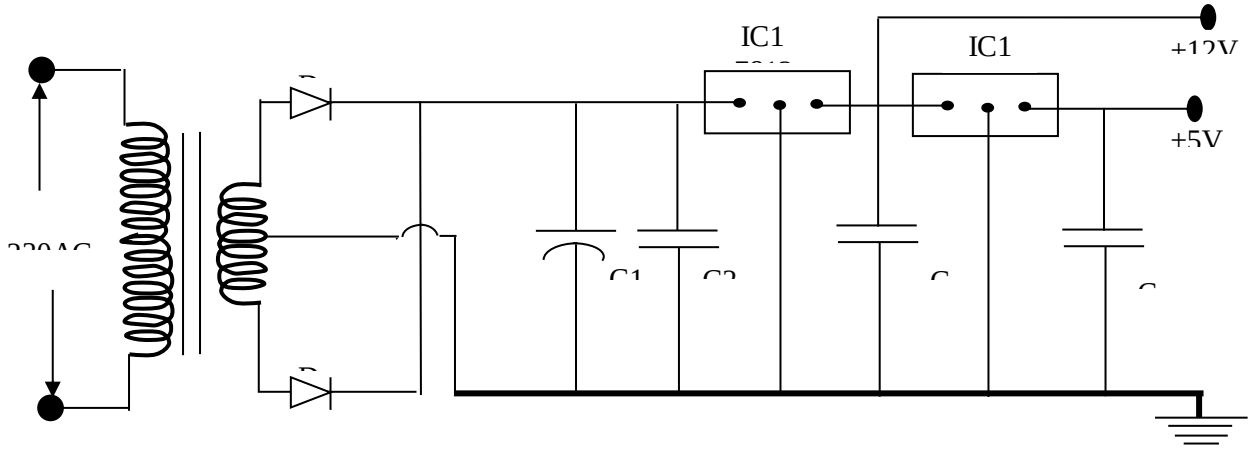


1 2 3

The stabilization of d.c. output is achieved by using the three terminal voltage regulator IC. This regulator IC comes in two flavors: 78xx for positive voltage output and 79xx for negative voltage output. For example 7805 gives +5V output and 7905 gives -5V stabilized output. These regulator ICs have in-built short-circuit protection and auto-thermal cutout provisions. If the load current is very high the IC needs 'heat sink' to dissipate the internally generated power.

Circuit Description: A d.c. power supply which maintains the output voltage constant irrespective of a.c. mains fluctuations or load variations is known as regulated d.c. power supply. It is also referred as full-wave regulated power supply as it uses four diodes in bridge fashion with the transformer. This laboratory power supply offers excellent line and load regulation and output voltages of +5V & +12 V at output currents up to one amp amp.

CIRCUIT DIAGRAM OF +5V & +12V FULL WAVE REGULATED POWER SUPPLY



Parts List:

SEMICONDUCTORS		
IC1	7812 Regulator IC	1
IC2	7805 Regulator IC	1
D1& D2	1N4007 Rectifier Diodes	2
CAPACITORS		
C1	1000 µf/25V Electrolytic	1
C2 to C4	0.1µF Ceramic Disc type	3
MISCELLANEOUS		
X1	230V AC Pri,14-0-14 1Amp Sec Transformer	1

1. Step-down Transformer: The transformer rating is 230V AC at Primary and 12-0-12V, 1Ampers across secondary winding. This transformer has a capability to deliver a current of 1Ampere, which is more than enough to drive any electronic circuit or varying load. The 12VAC appearing across the secondary is the RMS value of the waveform and the peak value would be 12 x 1.414 = 16.8 volts. This value limits our choice of rectifier diode as 1N4007, which is having PIV rating more than 16Volts.

2. Rectifier Stage: The two diodes D1 & D2 are connected across the secondary winding of the transformer as a full-wave rectifier. During the positive half-cycle of secondary voltage, the end A of the secondary winding becomes positive and end B negative. This makes the diode D1 forward biased and diode D2 reverse biased. Therefore diode D1 conducts while diode D2 does not. During the negative half-cycle, end A of the secondary winding becomes negative and end B positive. Therefore diode D2 conducts while diode D1 does not. Note that current across the centre tap terminal is in the same direction for both half-cycles of input a.c. voltage. Therefore, pulsating d.c. is obtained at point 'C' with respect to Ground.

3. Filter Stage: Here Capacitor C1 is used for filtering purpose and connected across the

rectifier output. It filters the a.c. components present in the rectified d.c. and gives steady d.c. voltage. As the rectifier voltage increases, it charges the capacitor and also supplies current to the load. When capacitor is charged to the peak value of the rectifier voltage, rectifier voltage starts to decrease. As the next voltage peak immediately recharges the capacitor, the discharge period is of very small duration. Due to this continuous charge-discharge-recharge cycle very little ripple is observed in the filtered output. Moreover, output voltage is higher as it remains substantially near the peak value of rectifier output voltage. This phenomenon is also explained in other form as: the shunt capacitor offers a low reactance path to the a.c. components of current and open circuit to d.c. component. During positive half cycle the capacitor stores energy in the form of electrostatic field. During negative half cycle, the filter capacitor releases stored energy to the load.

4. Voltage Regulation Stage: Across the point 'D' and Ground there is rectified and filtered d.c. In the present circuit KIA 7812 three terminal voltage regulator IC is used to get +12V and KIA 7805 voltage regulator IC is used to get +5V regulated d.c. output. In the three terminals, pin 1 is input i.e., rectified & filtered d.c. is connected to this pin. Pin 2 is common pin and is grounded. The pin 3 gives the stabilized d.c. output to the load. The circuit shows two more decoupling capacitors C2 & C3, which provides ground path to the high frequency noise signals. Across the point 'E' and 'F' with respect to ground +5V & +12V stabilized or regulated d.c output is measured, which can be connected to the required circuit.

Note: While connecting the diodes and electrolytic capacitors the polarities must be taken into consideration. The transformer's primary winding deals with 230V mains, care should be taken with it.

OVERVIEW OF THE SYSTEM

Our system aims to achieve the target to design a system that can provide following functionality-

Develop an android application that will act as an remote of a robot.

b. Develop a robot which will be helpful for travelling.

c. Here the focus is on the latest technology of android and robot also called as mobot. An android smartphone and the technology of android is vast and can be used to interact with embedded

system. Mobile, robot and Bluetooth are the on-going technologies which can be used for the benefit of mankind.

The system will consist of following four parts:

a. Bluetooth technology,
b. Android smartphone
c. Microcontroller and
d. DC motor.

Android smartphone will act as remote of the system, Bluetooth will act as the connection link between robot and android smartphone, microcontroller will act as the brain of the robot, and DC motor will help us to move the robot. This report describes about the real time android technology that can be used as an interaction between embedded system.

FUNCTIONALITY PROVIDED BY THE SYSTEM

Robot control design using a smartphone is the direct usage of a simple device that is as simple as our easy to carry, the mobile phones to drive a robot which is considered quiet complex when comparisons are made to the mobile phones. Here in this system the mobile phone will carry out the controlling of robot with the help of a smartphone application. The smartphone application will be developed in such a way that it is compatible to all the current versions of Android. The smartphone application will communicate to the robot with the help of Bluetooth
module which is fitted on the Robot. The robot that is the skateboard (from the view of our project) is mechanically modified in order to fit the Bluetooth module and microcontroller. The microcontroller will read the instructions from the smartphone application with the help of Bluetooth and gradually will control the Robot with the help of DC motor.

SYSTEM INTERFACES, INPUTS, AND OUTPUTS

The system interface consists of following entities-Smartphone device, Android Application, Skateboard (Robot), Microcontroller and DC Motor. The Smartphone device acts as the base hardware on which the Android Application plays the role of an interface. The input is given to the skateboard through the application which is fed to the skateboard through Bluetooth module which transfers the instructions passed by the smartphone application to the skateboard. The DC motor is also in

synchronization with the microcontroller which makes the robot perform the required kinds of motion.

EXPECTED OUTCOMES

The robot (skateboard) should be used for travelling as a basic outcome with minimal energy usage. Since the motors used are completely environment friendly, the travelling for shorter distances could be made feasible.

LCD INTERFACED IN 4 BIT MODE

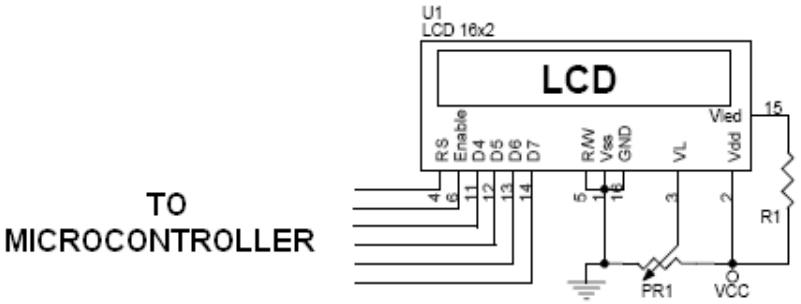


Fig (13)

LCD INTERFACED IN 8 BIT MODE

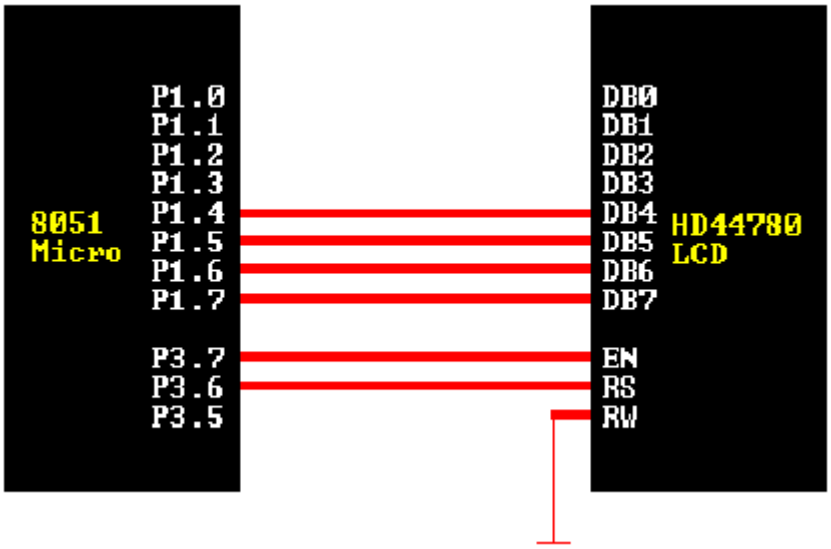


Fig (14)

UNDERSTANDING LCD

Pin out

• 8 data pins D7:D0

Bi-directional data/command pins.

Alphanumeric characters are sent in ASCII format.

• RS: Register Select

RS = 0 -> Command Register is selected

RS = 1 -> Data Register is selected

• R/W: Read or Write

0 -> Write, 1 -> Read

• E: Enable (Latch data)

Used to latch the data present on the data pins.

A high-to-low edge is needed to latch the data.

• VEE: contrast control

NOTE: When writing to the display, data is transferred only on the high to low transition of this signal. However, when reading from the display, data will become available shortly after the low to high transition and remain available until the signal falls low again.

Display Data RAM (DDRAM)

Display data RAM (DDRAM) is where you send the characters (ASCII code) you want to see on the LCD screen. It stores display data represented in 8-bit character codes. Its capacity is 80 characters (bytes).

In the above memory map, the area shaded in black is the visible display (For 16x2 display) .
For first line addresses for first 15 characters is from 00h to 0Fh. But for second line address of first character is 40h and so on up to 4Fh for the 16th character.

So if you want to display the text at specific positions of LCD, we require to manipulate address and then to set cursor position accordingly.

Character Generator RAM (CGRAM)-User defined character RAM

In the character generator RAM, we can define our own character patterns by program. CG RAM is 64 bytes, allowing for eight 5*8 pixel, character patterns to be defined. However how to define this and use it is out of scope of this tutorial. So I will not talk any more about CGRAM

Registers

The HD44780 has two 8-bit registers, an instruction register (IR) and a data register (DR). The IR stores instruction codes. The DR temporarily stores data to be written into DDRAM or CGRAM and temporarily stores

Data to be read from DDRAM or CGRAM. Data written into the DR is automatically written into DDRAM or CGRAM by an internal operation. . These two registers can be selected by the register selector (RS) signal. See the table below:

		Register Selection
RS	R/W	Operation
0	0	IR write as an internal operation (display clear, etc.)
0	1	Read busy flag (DB7) and address counter (DB0 to DB6)
1	0	DR write as an internal operation (DR to DDRAM or CGRAM)
1	1	DR read as an internal operation (DDRAM or CGRAM to DR)

When the busy flag is 1, the LCD is in the internal operation mode, and the next instruction will not be accepted. When RS = 0 and R/W = 1 (see the table above), the busy flag is output to DB7 (MSB of LCD data bus). The next instruction must be written after ensuring that the busy flag is 0.

Instruction	**Decimal**	**HEX**

Function set (8-bit interface, 2 lines, 5*7 Pixels)	56	38
Function set (8-bit interface, 1 line, 5*7 Pixels)	48	30
Function set (4-bit interface, 2 lines, 5*7 Pixels)	40	28
Function set (4-bit interface, 1 line, 5*7 Pixels)	32	20
Entry mode set	See Below	See Below
Scroll display one character right (all lines)	28	1E
Scroll display one character left (all lines)	24	18
Home (move cursor to top/left character position)	2	2
Move cursor one character left	16	10
Move cursor one character right	20	14
Turn on visible underline cursor	14	0E
Turn on visible blinking-block cursor	15	0F
Make cursor invisible	12	0C
Blank the display (without clearing)	8	08
Restore the display (with cursor hidden)	12	0C
Clear Screen	1	01
Set cursor position (DDRAM address)	128 + addr	80+ addr
Set pointer in character-generator RAM (CG RAM address)	64 + addr	40+ addr

Commands

The LCD's internal controller accepts several commands and modifies the display accordingly. These commands would be things like:

– Clear screen

– Return home– Shift display right/left

Entry mode set

This command sets cursor move direction and display shift ON/OFF. There are 4 possible function set commands; 04, 05, 06, and 07. This command changes the direction the cursor moves by setting the address counter to increment or decrement. This command is very important. If you do not understand it you may not see anything or what you actually wanted to see on LCD screen. I have created 4 animated gifs to demonstrate what the function set command is all about.

Set cursor position (DDRAM address)

As said earlier if we want to display the text at specific positions of LCD, we require to manipulate address and then to set cursor position accordingly.

I want to display "JINAY" in message "Hi JINAY" at the right corner of first line then I should start from 10th character.
So referring to table 80h+0Ah= 8Ah.

INTERFACING LCD TO 89C52

The 44780 standard requires 3 control lines as well as either 4 or 8 I/O lines for the data bus. The user may select whether the LCD is to operate with a 4-bit data bus or an 8-bit data bus. If a 4-bit data bus is used, the LCD will require a total of 7 data lines. If an 8-bit data bus is used, the LCD will require a total of 11 data lines. The three control lines are EN, **RS**, and **RW**.

Note that the EN line must be raised / lowered before/after each instruction sent to the LCD regardless of whether that instruction is read or write text or instruction. In short, you must always manipulate EN when communicating with the LCD. EN is the LCD's way of knowing that you are talking to it. If you don't raise/lower EN, the LCD doesn't know you're talking to it on the other lines.

Checking the Busy Flag

You can use subroutine for checking busy flag or just a big (and safe)delay.
Set R/W Pin of the LCD HIGH(read from the LCD)

1. Select the instruction register by setting RS pin LOW
2. Enable the LCD by Setting the enable pin HIGH
3. The most significant bit of the LCD data bus is the state of the busy flag (1=Busy,0=ready to accept instructions/data). The other bits hold the current value of the address counter.

If the LCD never come out from ”busy" status because of some problems, the program will ”hang," waiting for DB7 to go low. So in a real applications it would be wise to put some kind of time limit on the delay--for example, a maximum of 100 attempts to wait for the busy signal to go low. This would guarantee that even if the LCD hardware fails, the

program would not lock up.

Chapter 5

Result & Discussion

SIM800L GSM/GPRS Module

The SIM800L is a **low-cost GSM/GPRS module** designed for IoT tracking devices and embedded GPS solutions, making it ideal for DIY location monitoring projects. It operates on quad-band frequencies (**850, 900, 1800, and 1900 MHz**), making it compatible with **2G** networks worldwide. Its small form factor and UART-based communication make it ideal for microcontroller integration.

Key Features and Capabilities

⇒ **Voice, SMS, and GPRS Data:** The SIM800L can make and receive voice calls, send and receive SMS messages, and connect to the internet using GPRS. This makes it versatile for remote monitoring and alert-based systems.

⇒ **TCP/UDP and HTTP Support:** It supports TCP/IP stack and HTTP via AT commands, enabling interaction with web APIs or cloud databases.

⇒ AT Command Control: Controlled entirely via AT commands over UART. It responds to common commands like AT, AT+CSQ, AT+CREG?, AT+HTTPACTION, etc.

⇒ **Low Power Modes:** Offers various power-saving options, including sleep and idle states, for battery-powered projects.

⇒ **Network Functions:** Can scan for available networks, handle automatic or manual network registration, and report signal quality.

Hardware Considerations

⇒ **Voltage Sensitivity:** It must be powered with a stable supply of 3.7V to 4.2V, and it can draw up to 2A of current during transmission bursts.

⇒ **Antenna:** Requires an external GSM antenna for proper signal strength. Antenna types include PCB, helical, or SMA-based external ones.

⇒ **Logic Levels:** Operates at 2.8V to 3V logic level, but in our testing, it was able to withstand 3.3V on MCU Pins. Use a level shifter if you are using a standalone SIM800L IC.

Neo 6M GPS Module with Raspberry Pi

The **Neo-6M** is a compact and cost-effective GPS receiver module developed by u-blox, designed specifically for tracking and navigation applications. It supports only the GPS (Global Positioning System) constellation. The setup() function initialises both UART interfaces for the Neo 6M GPS module with Raspberry Pi communication and SIM800L cellular connectivity.

The module is popular for its low power consumption, high positioning accuracy, and fast time-to-first-fix (TTFF). It communicates using standard NMEA 0183 sentences over a UART interface, typically at a default baud rate of 9600 bps. Due to its reliability and affordability, it's widely used in vehicle tracking systems, drones, robotics, and DIY navigation devices.

Key Features and Capabilities

⇒ **NMEA 0183 Protocol Output:** Continuously transmits GPS data such as time, latitude, longitude, altitude, number of satellites, speed, and fix quality via standard NMEA sentences like $GPGGA, $GPRMC, $GPGSV, and others.

⇒ **High Sensitivity Receiver:** Capable of acquiring weak satellite signals (-161 dBm tracking sensitivity), making it suitable for outdoor use with minimal obstruction.

⇒ **Fast Time-to-First-Fix (TTFF):** Supports hot, warm, and cold starts with TTFF as low as 1 second (hot) and 27 seconds (cold).

⇒ **Position Accuracy:** Offers 2.5m CEP positioning accuracy under clear sky conditions.

⇒ **Backup Battery Support:** Can be equipped with a coin cell to retain satellite data and RTC for faster startups.

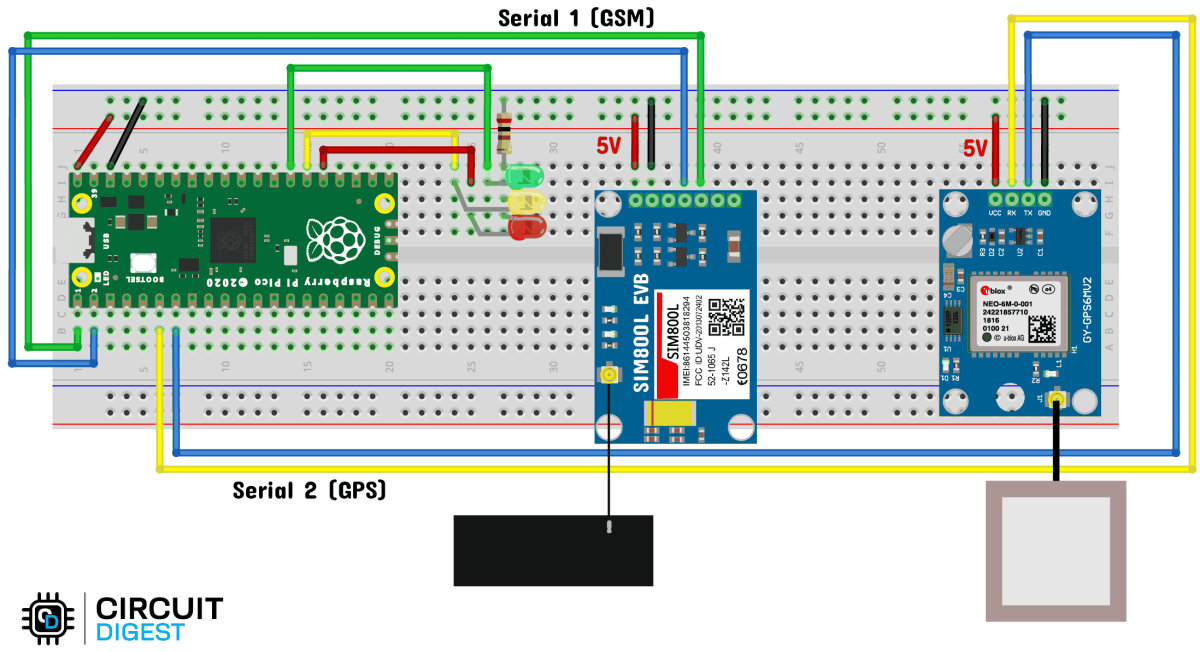
Hardware Considerations

⇒ **Antenna Requirements:** Works best with an active GPS antenna for improved signal acquisition. Typically includes a UFL or SMA connector.

⇒ **Power Supply:** IC Operates at 3.3V and draws around 45–50 mA. Avoid powering from 5V directly if your module doesn't have a power regulator(LDO).

⇒ **Status LED:** Blinks when the module has acquired a valid fix, providing easy visual feedback.

Raspberry Pi Pico GPS Tracker Circuit Diagram & Wiring Guide



SIM800L Pins	Pico GPIO Pins	Description
TX	GP1 (RX)	Receive from SIM800L
RX	GP0 (TX)	Send to SIM800L
GND	GND	Common ground
VCC	Power	External battery or supply (3.7V - 4.2V)

Neo-6M Pins	Pico GPIO Pins	Description
TX	GP5 (RX)	Receive GPS data
RX	GP4 (TX)	
GND	GND	Common ground
VCC	Power	External battery or supply (3.6V)

Note: Ensure that both the SIM800L and Neo-6M share a common ground with the Pico. Also, if your SIM800L & Neo-6M Modules have an inbuilt regulator(LDO), you can connect VCC directly to the 5V or VBUS pin in Pico.

What is GSM?

Mobile Communication System - Global system for mobile communication - GSM - is a wide area wireless communications system that uses digital radio transmission to provide voice, data, and multimedia communication services. A GSM system coordinates the communication between mobile telephones (mobile stations), base stations (cell sites), and switching systems.

Digital Media Formats - GSM is designed to transfer digital information. The initial version of GSM transmitted digital media in **circuit switched** (continuous transmission) form and later versions of GSM deliver data in **packet data** form.

Functional Sections - GSM system is composed of three key sections:

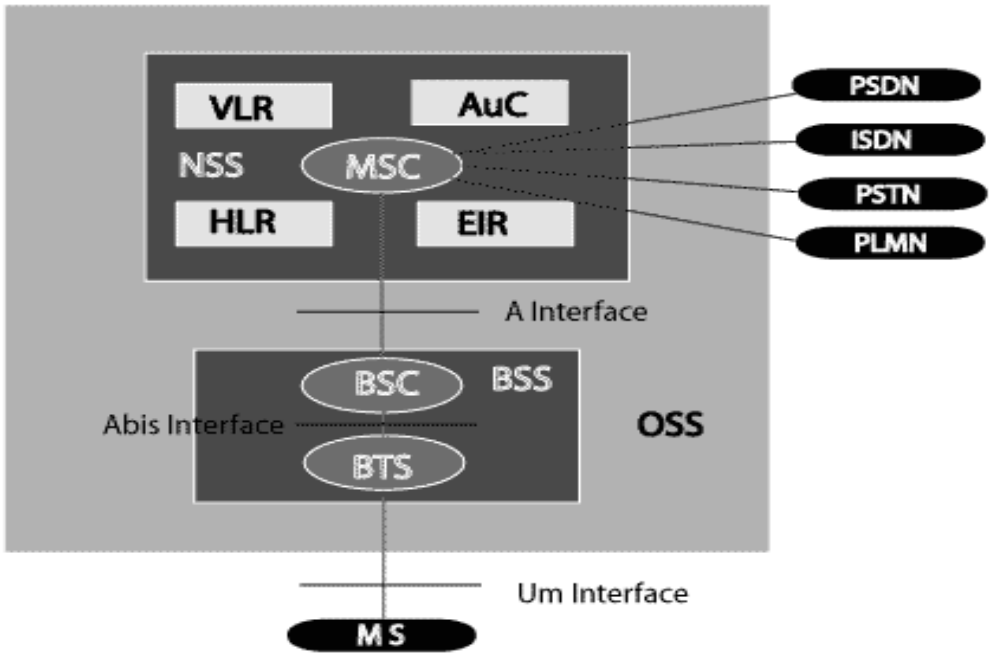
1. **Mobile Stations (MS)** - A device that converts media to and from GSM radio signals.

2. **Base Station Subsystem (BSS)** - Assemblies that convert digital signals to radio signals that can be sent to mobile devices and receive radio signals that can be converted back to their digital form. The BSS is divided into **base station** - **BS** - parts that are located at the cell site and **base station controllers** - **BSC** - that coordinate the distribution and reception of communication connections
3. **Network and Switching System (NSS)** - The NSS performs the **interconnection** between the **base station** parts and **other networks** such as the public switched telephone network - **PSTN** - and public**Internet**. The NSS is composed of circuit data and packet data switches, databases, and administrative control services.

   A GSM network consists of several functional entities, whose functions and interfaces are defined. The GSM network can be divided into following broad parts.

- The Mobile Station (MS)
- The Base Station Subsystem (BSS)
- The Network Switching Subsystem (NSS)
- The Operation Support Subsystem (OSS)

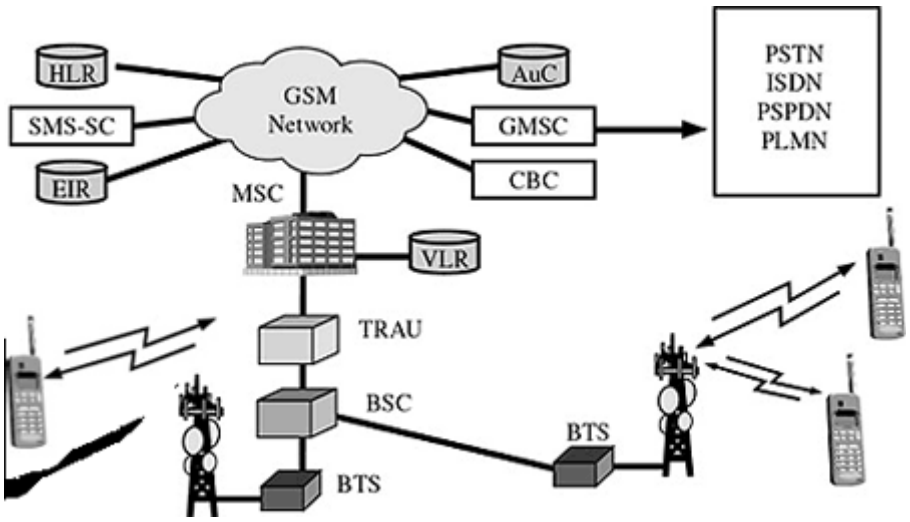
Following is the simple architecture diagram of GSM Network:



The added components of the GSM architecture include the functions of the databases and messaging systems:

- Home Location Register (HLR)
- Visitor Location Register (VLR)
- Equipment Identity Register (EIR)
- Authentication Center (AuC)
- SMS Serving Center (SMS SC)
- Gateway MSC (GMSC)
- Chargeback Center (CBC)
- Transcoder and Adaptation Unit (TRAU)

Following is the diagram of GSM Netwrok along with added elements:



The MS and the BSS communicate across the Um interface, also known as the air interface or radio link. The BSS communicates with the Network Service Switching center across the A interface.

GSM network areas:

In a GSM network, the following areas are defined:

- **Cell:** Cell is the basic service area: one BTS covers one cell. Each cell is given a Cell Global Identity (CGI), a number that uniquely identifies the cell.

- **Location Area:** A group of cells form a Location Area. This is the area that is paged when a subscriber gets an incoming call. Each Location Area is assigned a Location Area Identity (LAI). Each Location Area is served by one or more BSCs.
- **MSC/VLR Service Area:** The area covered by one MSC is called the MSC/VLR service area.
- **PLMN:** The area covered by one network operator is called PLMN. A PLMN can contain one or more MSCs.

  - GPS

The Global Positioning System (GPS) is a satellite-based navigation system made up of a network of 24 satellites placed into orbit by the U.S. Department of Defence. Military actions was the original intent for GPS, however in the 1980s, the U.S. government decided to allow the GPS program to be used by civilians. Weather conditions do not affect the ability for GPS to work. The systems work 24/7 anywhere in the world. There are no subscription fees or setup charges to use GPS.



GPS devices can have capabilities such as:

- Maps, including streets maps, displayed in human readable format via text or in a graphical format,
- Turn-by-turn navigation directions to a human in charge of a vehicle or vessel via text or speech,
- Directions fed directly to an autonomous vehicle such as a robotic probe,
- Traffic congestion maps (depicting either historical or real time data) and suggested alternative directions,
- Information on nearby amenities such as restaurants, fueling stations, and tourist attractions.

GPS can able to answer:

- The roads or paths available,
- Traffic congestion and alternative routes,
- Roads or paths that might be taken to get to the destination,
- If some roads are busy (now or historically) the best route to take,
- The location of food, banks, hotels, fuel, airports or other places of interests,
- The shortest route between the two locations,
- The different options to drive on highway or back roads.

Of all the applications of GPS, vehicle tracking and navigational systems have brought this technology to the day-to-day life of the common man. Today GPS fitted cars; ambulances, fleets and police vehicles are common sights on the roads of developed countries. Known by many names such as Automatic Vehicle Locating System (AVLS), Vehicle Tracking and Information System (VTIS), Mobile Asset Management System (MAMS), these systems offer an effective tool for improving the operational efficiency and utilization of vehicles.

GPS is used in vehicles for both tracking and navigation. Tracking systems enable a base station to keep track of the vehicles without the intervention of the driver where, as navigation system helps the driver to reach the destination. Whether navigation system or tracking system, the architecture is more or less similar. The navigation system will have convenient, usually a graphic, display for the driver which is not needed for a tracking system. Vehicle Tracking Systems combine a number of well-developed technologies. Irrespective of the technology being used, VTS consist of three subsystems: a) In-vehicle unit (IVU), b) Base station and c) Communication link. The IVU includes a suitable position sensor and an intelligent controller together with an appropriate interface to the communication link. Thanks to the US Government announcement of 911E regulation, radio based position technology has witnessed a spurt of developmental activities.

Network Overlay Systems use cell phone infrastructure for locating vehicles. The cell centers with additional hardware and software assess the time of arrival (TOA) and angle of arrival (AOA) of radio signals from vehicles to compute the position of the vehicles. This information is sent to the tracking centre through the cell link or conventional link. Another technique used for locating vehicles computes the time difference for signals from

two cell centers to reach the vehicle. This computation is made in the IVU and the position information is sent to the tracking centre through the cell phone link. A more common technique used is direct radio link (DRL). In this system dedicated radio infrastructure is used along with special IVU to compute vehicle location. However all these techniques impose limitation on the operational area. Alternatively, embedded GPS receivers provide absolute position co-ordinates at any point, without any area restrictions.

Mobile phones with GPS capability:

Due in part to regulations encouraging mobile phone tracking, including E911, the majority of GPS receivers are built into mobile telephones, with varying degrees of coverage and user accessibility. Commercial navigation software is available for most 21st-century smart phones as well as some Java-enabled phones that allows them to use an internal or external GPS receiver (in the latter case, connecting via serial or Bluetooth). Some phones using assisted GPS (A-GPS) function poorly when out of range of their carrier's cell towers. Others can navigate worldwide with satellite GPS signals as well as a dedicated portable GPS receiver does, upgrading their operation to A-GPS mode when in range. Still others have a hybrid positioning system that can use other signals when GPS signals are inadequate.

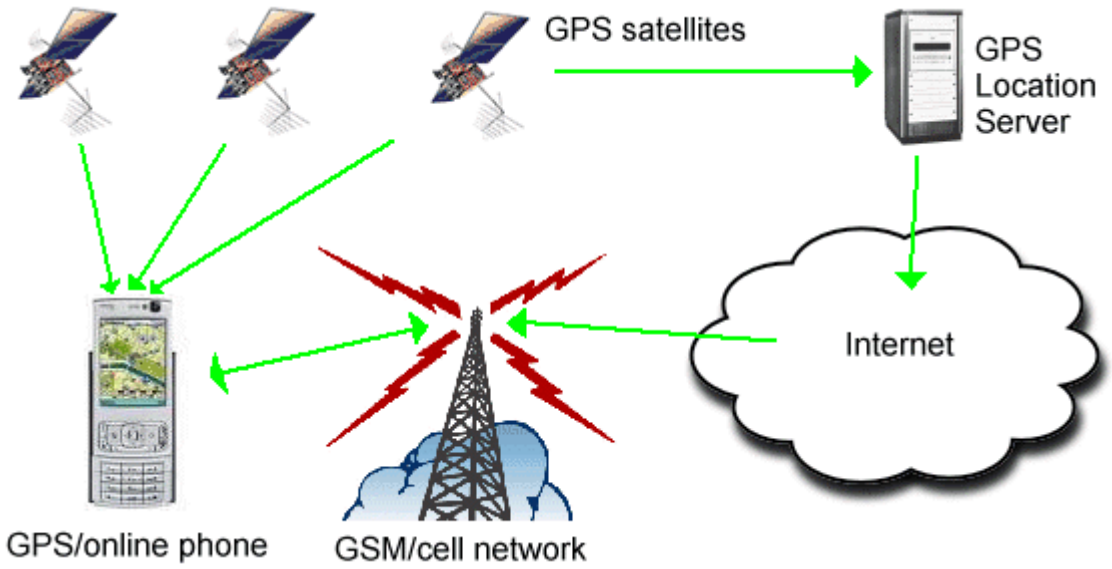
More bespoke solutions also exist for smart phones with inbuilt GPS capabilities. Some such phones can use tethering to double as a wireless modem for a laptop, while allowing GPS-navigation/localisation as well. One such example is marketed by Verizon Wireless in the United States, and is called VZ Navigator. The system uses gps One technology to determine the location, and then uses the mobile phone's data connection to download maps and calculate navigational routes. Other products including iPhone are used to provide similar services. Nokia gives Ovi Maps free on its smart phones and maps can be preloaded.

According to market research from the independent analyst firm Berg Insight, the sales of GPS-enabled GSM/WCDMA handsets was 150 million units in 2009,[4] while only 40 million separate GPS receivers were sold.

GPS navigation applications for mobile phones include on-line (e.g. Waze, Google Maps Navigation) and off-line (e.g. iGo for Android, Maverick and Here (Nokia) for Windows Phone) navigation applications. Google Maps Navigation, which is included with Android,

means most smart phone users only need their phone to have a personal navigation assistant.

Many Android smart phones have an additional GPS feature, called **EPO** (Extended Prediction Orbit). The phone downloads a file to help it locate GPS satellites more quickly and reduce the Time To First Fix.



Assisted GPS, as the Nokia N95, speeds up the TTFF significantly in two ways. The received GPS signals are shifted in frequency due to the relative receiver-satellite motion (Doppler shift). The GPS receiver must find the frequency of the signal before it can lock on. Knowledge of each satellite's position and velocity from an Assisted GPS server effectively reduces the number of frequencies to be searched because there's less guesswork involved, speeding TTFF up by (potentially) tens of seconds.

Even more significantly, normally a receiver has to spend at least 20 seconds (and often a lot more) decoding the navigation data from the satellites it can find, before it can start the triangulation calculations to work out a position on the Earth's surface. With Assisted GPS, the navigation data is provided, more or less perfectly (think of it as a crib that the GPS can use).

Add the two factors together and it's easy to see that even in worst case, TTFF can be reduced by 30 seconds or so. A final benefit comes from increased receiver sensitivity, which is directly related to the number of frequencies which must be searched to find a satellite signal. Because the GPS receiver has fewer frequencies to search, it can dwell on

each for a greater period of time, increasing the effective sensitivity and enabling the GPS to use signal strengths below the normal thresholds to make range measurements (for example in built-up areas where many satellites are blocked by obstructions).

At the moment, management of GPS positioning in smartphones is semi-automatic but there's often still a little fiddling to be done by the end user. Once the technology goes mainstream though, I'd expect a fully automatic solution, integrated into the smartphone OS. Ideally, we'd see Assisted GPS use AND intelligent power control of the GPS to only require fixes when position-dependent applications were active and when movement is detected within the GSM cell (to save power). Once this is in place, we get to the situation where every smartphone will effectively know where it is on the planet at all times, with perhaps only a brief (10 seconds or so) 'Please wait' message every now and then when a very exact GPS fix is needed for navigation purposes.

The obvious application of GPS services is for providing maps of where you are and directions to get to somewhere else, an area which is well covered at the moment but which will only get more and more strong

ATMEGA328P is high performance, low power controller from Microchip. ATMEGA328P is an 8-bit microcontroller based on AVR RISC architecture. It is the most popular of all AVR controllers as it is used in ARDUINO boards.

ATMega328 Pin Configuration

ATMEGA328P is a 28 pin chip as shown in pin diagram above. Many pins of the chip here have more than one function. We will describe functions of each pin in below table.

Pin No.	Pin name	Description	Secondary Function
1	PC6 (RESET)	Pin6 of PORTC	Pin by default is used as RESET pin. PC6 can only be used as I/O pin when RSTDISBL Fuse is programmed.
2	PD0 (RXD)	Pin0 of PORTD	RXD (Data Input Pin for USART)

			USART Serial Communication Interface [Can be used for programming]
3	PD1 (TXD)	Pin1 of PORTD	TXD (Data Output Pin for USART) USART Serial Communication Interface [Can be used for programming] INT2(External Interrupt 2 Input)
4	PD2 (INT0)	Pin2 of PORTD	External Interrupt source 0
5	PD3 (INT1/OC2B)	Pin3 of PORTD	External Interrupt source1 OC2B(PWM - Timer/Counter2 Output Compare Match B Output)
6	PD4 (XCK/T0)	Pin4 of PORTD	T0(Timer0 External Counter Input) XCK (USART External Clock I/O)
7	VCC		Connected to positive voltage
8	GND		Connected to ground
9	PB6 (XTAL1/TOSC1)	Pin6 of PORTB	XTAL1 (Chip Clock Oscillator pin 1 or External clock input) TOSC1 (Timer Oscillator pin 1)
10	PB7 (XTAL2/TOSC2)	Pin7 of PORTB	XTAL2 (Chip Clock Oscillator pin 2) TOSC2 (Timer Oscillator pin 2)
11	PD5 (T1/OC0B)	Pin5 of PORTD	T1(Timer1 External Counter Input)

			OC0B(PWM - Timer/Counter0 Output Compare Match B Output)
12	PD6 (AIN0/OC0A)	Pin6 of PORTD	AIN0(Analog Comparator Positive I/P) OC0A(PWM - Timer/Counter0 Output Compare Match A Output)
13	PD7 (AIN1)	Pin7 of PORTD	AIN1(Analog Comparator Negative I/P)
14	PB0 (ICP1/CLKO)	Pin0 of PORTB	ICP1(Timer/Counter1 Input Capture Pin) CLKO (Divided System Clock. The divided system clock can be output on the PB0 pin)
15	PB1 (OC1A)	Pin1 of PORTB	OC1A (Timer/Counter1 Output Compare Match A Output)
16	PB2 (SS/OC1B)	Pin2 of PORTB	SS (SPI Slave Select Input). This pin is low when controller acts as slave. [Serial Peripheral Interface (SPI) for programming] OC1B (Timer/Counter1 Output Compare Match B Output)
17	PB3 (MOSI/OC2A)	Pin3 of PORTB	MOSI (Master Output Slave Input). When controller acts as slave, the data is received by this pin. [Serial Peripheral Interface (SPI) for programming] OC2 (Timer/Counter2 Output Compare Match Output)
18	PB4 (MISO)	Pin4 of PORTB	MISO (Master Input Slave Output). When controller acts as slave, the data is sent to master by this controller through this pin.

			[Serial Peripheral Interface (SPI) for programming]
19	PB5 (SCK)	Pin5 of PORTB	SCK (SPI Bus Serial Clock). This is the clock shared between this controller and other system for accurate data transfer. [Serial Peripheral Interface (SPI) for programming]
20	AVCC		Power for Internal ADC Converter
21	AREF		Analog Reference Pin for ADC
22	GND		GROUND
23	PC0 (ADC0)	Pin0 of PORTC	ADC0 (ADC Input Channel 0)
24	PC1 (ADC1)	Pin1 of PORTC	ADC1 (ADC Input Channel 1)
25	PC2 (ADC2)	Pin2 of PORTC	ADC2 (ADC Input Channel 2)
26	PC3 (ADC3)	Pin3 of PORTC	ADC3 (ADC Input Channel 3)
27	PC4 (ADC4/SDA)	Pin4 of PORTC	ADC4 (ADC Input Channel 4) SDA (Two-wire Serial Bus Data Input/output Line)
28	PC5 (ADC5/SCL)	Pin5 of PORTC	ADC5 (ADC Input Channel 5) SCL (Two-wire Serial Bus Clock Line)

Features

ATMEGA328P – Simplified Features

CPU	8-bit AVR
Number of Pins	28
Operating Voltage (V)	+1.8 V TO +5.5V
Number of programmable I/O lines	23
Communication Interface	Master/Slave SPI Serial Interface(17,18,19 PINS) [Can be used for programming this controller] Programmable Serial USART(2,3 PINS) [Can be used for programming this controller] Two-wire Serial Interface(27,28 PINS)[Can be used to connect peripheral devices like Servos, sensors and memory devices]
JTAG Interface	Not available
ADC Module	6channels, 10-bit resolution ADC
Timer Module	Two 8-bit counters with Separate Prescaler and compare mode, One 16-bit counter with Separate Prescaler,compare mode and capture mode.
Analog Comparators	1(12,13 PINS)
DAC Module	Nil
PWM channels	6
External Oscillator	0-4MHz @ 1.8V to 5.5V 0-10MHz @ 2.7V to 5.5V 0-20MHz @ 4.5V to 5.5V

Internal Oscillator	8MHz Calibrated Internal Oscillator
Program Memory Type	Flash
Program Memory or Flash memory	32Kbytes[10000 write/erase cycles]
CPU Speed	1MIPS for 1MHz
RAM	2Kbytes Internal SRAM
EEPROM	1Kbytes EEPROM
Watchdog Timer	Programmable Watchdog Timer with Separate On-chipOscillator
Program Lock	Yes
Power Save Modes	Six Modes[Idle, ADC Noise Reduction, Power-save, Power-down, Standby and Extended Standby]
Operating Temperature	-40℃ to +105℃(+105 being absolute maximum, -40 being absolute minimum)

What is Raspberry Pi Pico? If you are a young maker, chances are you already know about the Raspberry Pi. The Raspberry Pi Pico then is the latest offering from the manufacturers of the Raspberry Pi. In this blog, we shall take a look at Raspberry Pi Pico, its key features (including hardware and software specifications), its pinout, how one can set it up and even light an LED using it.

What is Raspberry Pi Pico?

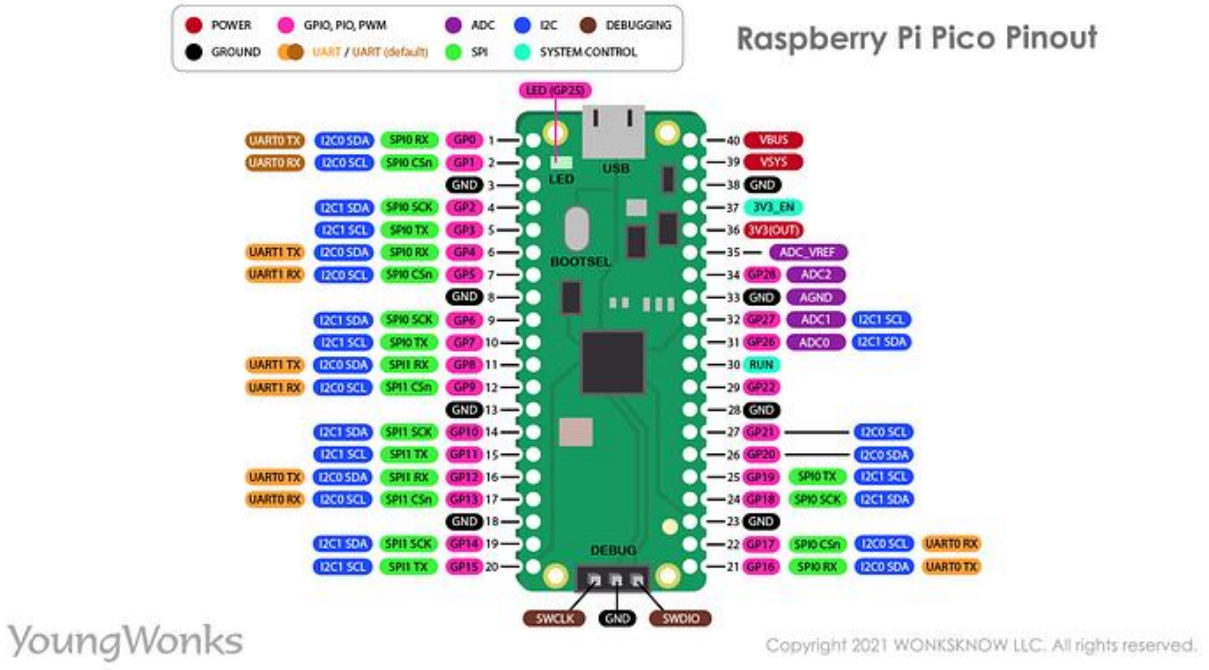
Raspberry Pi refers to a series of small single-board computers developed in the United Kingdom by the Raspberry Pi Foundation in association with Broadcom. The boards have been made keeping in mind the promotion of teaching basic computer science to kids. The latest offering by Raspberry Pi is the Raspberry Pi Pico, a new flexible IoT board. Essentially, it is a microcontroller board built on silicon and designed at the Raspberry Pi Foundation.

Priced at merely $4, Raspberry Pi Pico is smaller than the average Pi and indicates how the foundation is now looking to branch out into microcontrollers and custom silicon.

Raspberry Pi Pico pinout

In order to use it well, it is recommended to get well acquainted with the Raspberry Pi Pico pinout.

The diagram below explains this.
Press enter or click to view image in full size



Raspberry Pi Pico projects

As a microcontroller, the Raspberry Pi Pico can be used in many projects, be it Internet of Things (IoT), Adafruit Neopixel projects, data logging, small to medium scale robotics, projects needing interfacing with cameras, analog sensing (using environment sensors) and more.

Chapter 6

Future Scope & Conclusion

6.1. Future Scope

The project can be improved further by enhancing autonomous navigation, sensor integration, communication reliability, power management, mechanical design and software performance. Future development should be carried out in a controlled testing environment, with emphasis on reliability, operator safety and responsible use.

6.1.1 Improved Autonomous Navigation

The robot can be developed to navigate more reliably through unfamiliar indoor or outdoor environments. ROS 2 navigation packages, a correctly configured map, localization and obstacle-detection inputs can be integrated to support waypoint navigation. Future testing can evaluate how consistently the robot follows a planned route, stops for obstacles and recovers when its path is blocked. Performance should be measured across different surfaces and lighting or environmental conditions.

6.1.2 Better Mapping and Localization

The mapping system can be improved by combining LiDAR data with wheel-encoder odometry and IMU measurements. Where outdoor operation is required, GPS/GNSS data may also be integrated. Sensor fusion can help reduce the effect of errors from any one sensor, but it requires calibration, correct coordinate frames and careful configuration. Future work can compare the estimated position with a known reference to evaluate localization accuracy.

6.1.3 Improved Camera and Computer-Vision Capability

The camera can be integrated with OpenCV and a suitable object-detection model such as YOLO for visual recognition and tracking in permitted test scenarios. Future work may improve image quality, camera mounting, frame processing speed and performance under changing lighting. Any detection result should be evaluated using a test dataset and should not be treated as perfectly accurate. Camera-based results can be combined with distance measurements when a reliable, calibrated method is available.

6.1.4 More Reliable Motor Control

The motor-control system can be improved by using encoder feedback to regulate wheel speed and movement. A closed-loop controller can compare the requested speed with measured speed and adjust motor commands to reduce error. Future tests can measure speed consistency, turning accuracy, response to changes in load and behaviour on different

surfaces. The motor driver must be selected based on the actual motor current, including startup and stall conditions, and must have adequate thermal management.

6.1.5 Improved Communication and Remote Monitoring

The communication system can be developed to transmit useful status information such as battery level, connection state, sensor health and operating mode to a monitoring station. Wi-Fi may be appropriate in a local network, while cellular communication depends on network coverage and compatible hardware. Future work should include authentication, secure communication, clear loss-of-connection behaviour and an independent physical emergency stop. The robot should enter a safe state if communication becomes unreliable.

6.1.6 Power Management and Operating Time

Future development can measure the actual power consumption of the Raspberry Pi, ESP32, sensors, servos and motors. Battery capacity and expected operating time can then be estimated from measured loads rather than assumptions. Suitable voltage regulators, fuses, battery monitoring and a correctly rated battery-management system can improve reliability. Power tests should verify that motor loads do not cause the computer or sensors to reset due to voltage drops.

6.1.7 Software Integration and Diagnostics

The software can be organized into separate ROS 2 nodes and launch files so that sensors, localization, mapping, visualization and control can be started and tested systematically. Logging and diagnostic messages can help identify sensor disconnections, software failures or unusual motor behaviour. Future work can include repeatable startup procedures, parameter files, automated checks and documentation of software versions.

6.1.8 Mechanical and Environmental Improvements

The robot's frame, sensor mounts, wiring and cable protection can be reviewed to improve stability and serviceability. Sensors should be mounted rigidly and positioned so that the robot's own structure does not unnecessarily block their field of view. Future versions can be tested for vibration, dust exposure and surface irregularities appropriate to the intended operating environment. Changes should not be made without checking weight distribution, motor torque and the effect on sensor measurements.

6.1.9 Testing, Validation and Performance Measurement

A future testing plan can measure mapping quality, localization error, obstacle-detection performance, route-completion rate, motor-speed error, battery operating time and communication reliability. Tests should be repeated under documented conditions so that results can be compared between software or hardware revisions. A controlled test area, speed limits, a physical emergency stop and supervised testing are essential during development.

6.1.10 Expandability and Modular Design

The system can be designed so that individual sensors or software packages can be replaced without redesigning the entire robot. Standard ROS 2 interfaces, documented messages, clearly defined electrical connections and separate control modules can make future upgrades easier. Any added hardware should be checked for power, compute, bandwidth and software compatibility before integration.

6.2. Conclusion

The project, "Development of Autonomous Mobile Robot for Warfare," brings together computing, sensing, communication and motor-control components to form a mobile robotic platform. The Raspberry Pi 4 is intended to provide high-level processing and run the robotics software, while the ESP32 can perform low-level control tasks. LiDAR can provide distance measurements for mapping and obstacle detection; the camera can provide visual data; GPS and IMU can contribute positioning and motion information; and Johnson motors with encoders can provide movement and feedback. A suitable motor driver connects the controller to the motor power circuit, while the battery supplies electrical energy through appropriate power distribution.

Ubuntu can provide the operating-system environment, and ROS 2 can support communication between the software components. With compatible drivers and correct configuration, additional packages can be used for mapping, localization, navigation and visualization. These capabilities depend on the actual hardware, software integration, calibration and validation; installing Ubuntu or ROS 2 alone does not make the robot autonomous.

The project provides a platform for learning how embedded systems, sensors, robotics software and motor control can be integrated. It also highlights the importance of reliable

power design, correct sensor calibration, safe control behaviour and systematic testing. Further work is needed to validate navigation performance, communication reliability, battery operating time and the complete hardware configuration under representative conditions.

In conclusion, the proposed system offers a foundation for further development of an autonomous mobile robot. Its effectiveness should be established through measured results and controlled trials, with a focus on safe operation, reliability, maintainability and responsible use.

6.3. Expected Outcomes of Future Development

- More consistent navigation and obstacle response after mapping and navigation software are integrated and tested.
- Improved movement estimates through calibrated encoder, IMU and, where appropriate, GPS data fusion.
- More useful visual information through a tested camera-processing pipeline.
- Better monitoring of power, sensor status and communication health.
- Easier maintenance through modular software, clear wiring documentation and repeatable test procedures.
- Documented performance measurements that demonstrate the system's actual capabilities and remaining limitations.

Chapter 7

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✓ **Final Prototype**

